Develop a multithreaded counter application using POSIX threads (pthread_create, pthread_join). Demonstrate race conditions by allowing multiple threads to update a shared variable concurrently.

Modify the previous program using mutex locks (pthread_mutex_lock, pthread_mutex_unlock) and compare the outputs with the race-condition version. Analyze the impact of synchronization.

```
Last login: Sat Sep 26 22:59:02 on ttys000
varnikakomali@VARNIKAs-MacBook-Air ~ % cd OS_Lab
varnikakomali@VARNIKAs-MacBook-Air OS_Lab % cd Practical
varnikakomali@VARNIKAs-MacBook-Air Practical % nano practical_11.c
varnikakomali@VARNIKAs-MacBook-Air Practical % gcc practical_11.c -o practical_11 && ./practical_11
=== Part 1: Race Condition (Unsynchronized Counter) ===
Expected Value: 4000000
Actual Value:   1015171
Difference:     2984829 (Lost Updates)

=== Part 2: Synchronized Counter (POSIX Mutex Lock) ===
Expected Value: 4000000
Actual Value:   4000000
Difference:     0 (Thread-Safe Execution)
varnikakomali@VARNIKAs-MacBook-Air Practical %
```